U.S. Energy Generation
(2014-2024)

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ALY 3105 - Intermediate Statistics for Data Analytics
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Research Objectives

Evolution of Energy Sources

Examine the historical Changes and development of various energy sources over time.

Trends in Energy Shift

Identify and analyze discernable patterns in the transition from fossil fuels to renewable energy.

Statistical Significance

Determine if the observed relationships and trends in energy source evolution are statistically significance

Methods
Quantifying Trends Over Time

Descriptive Statistics

Summarize key data characteristics such as mean, median, and standard deviation.

Data Visualization

Utilize time series plots to identify trends, seasonality, and outliers.

Linear Regression Models

Quantify the linear trend within the time series data.

Dataset overview

Observations

11

Variables

6

Period

2014-2024

Data Quality

Clean, structured dataset

Energy Source Trends
An Initial Observation

Coal

Declining due to environmental regulations and competition from cleaner alternatives

Natural Gas

Increasing, often serving as a transitional fuel influenced by cost and availability

Solar

Experiencinggrowth fueled by technological advancements and cost reductions

Nuclear and Hydro

Remaining stable, providing consistent baseload and renewable power.

Energy Mix Transformation Shifting Towards Cleaner Energy Sources

The Transition

Moving from coal dependance to natural gas and solar power.

Current Status

Clean energy is growing, but fossil fuels stay a significant part of the supply.

Implication

Continued investment and policy support are critical for a fully clean energy future.

Fossil vs. Renewable Energy Trends

- **Fossil Fuels**

Historically leader, forming most of the energy consumption. While significant, their share is gradually declining due to environmental concerns and advancements in alternative cleaner energy.

- **Renewable Energy**

Experiencing exponential growth, influenced by technological advancements, falling costs, and supportive policies. The gap between fossil fuel and renewable energy is narrowing as renewables capture an increasing percentage of new energy capacity and investment.

Linear Regression: Purpose

- **Measure Trend Direction and Strength**

Quantify the relationship between variables to understand how changes in one variable impact another.

- **Identify Statistically Significant Changes**

Determine if observed relationships are likely real or due to random chance, enabling reliable conclusions.

Regression Analysis: Key Findings

- **Coal**

Demonstrates a strong negative trend.

- **Natural Gas**

Shows a positive trend.

- **Solar**

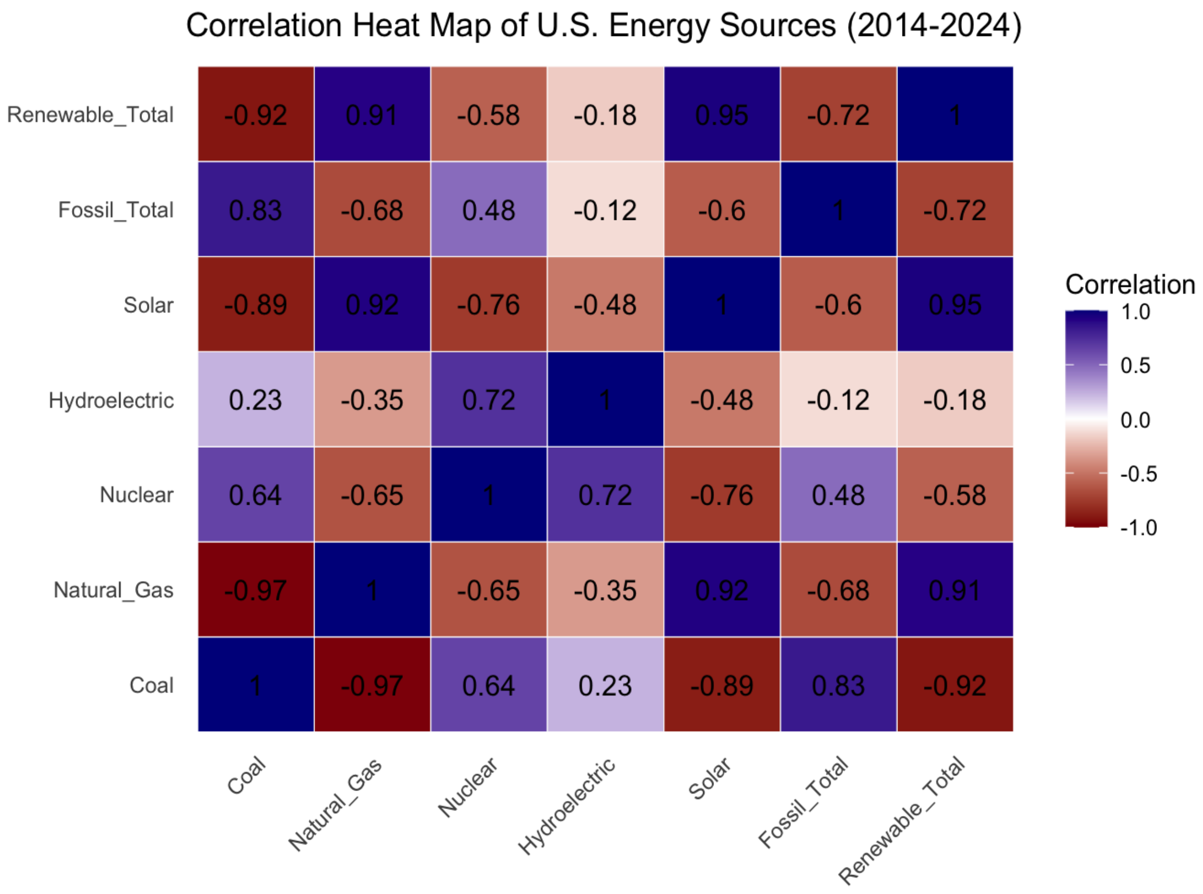
Exhibits the strongest positive growth.

- **Renewable Total**

Increasing significantly.

Heatmap Insight

- Coal negatively correlates with Solar and Renewable Total.
- Natural Gas positively increases while Coal declines.
- Renewable Total strongly aligns with Solar growth.



Key Insights on the Energy Transition

- **Gradual Shift**

The transition to cleaner energy sources is a progressive, long-term process, not immediate.

- **Structural Change Required**

Significant systematic adjustments in infrastructure, policy, and market dynamics are essential to accelerate and sustain the energy transition.

- **Natural Gas as a Bridge**

Natural gas continues to be a transitional fuel, balancing energy needs during the shift from traditional fossil fuels to renewables.

Conclusion
Key Findings

- The energy transition is actively underway.
- However, the process is far from complete.
- Fossil Fuels continue to be a primary source of energy in the U.S.

What I Learned

- **Time Series Data Analysis**

Proficiency in analyzing sequential data collected over time.

- **Regression Models for Predictive Insights**

Applying statistical models to understand relationships.

- **Interpretation of Energy Trends**

Ability to understand and explain real-world patterns and behavior in energy data.

Bibliography

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Thank you!

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